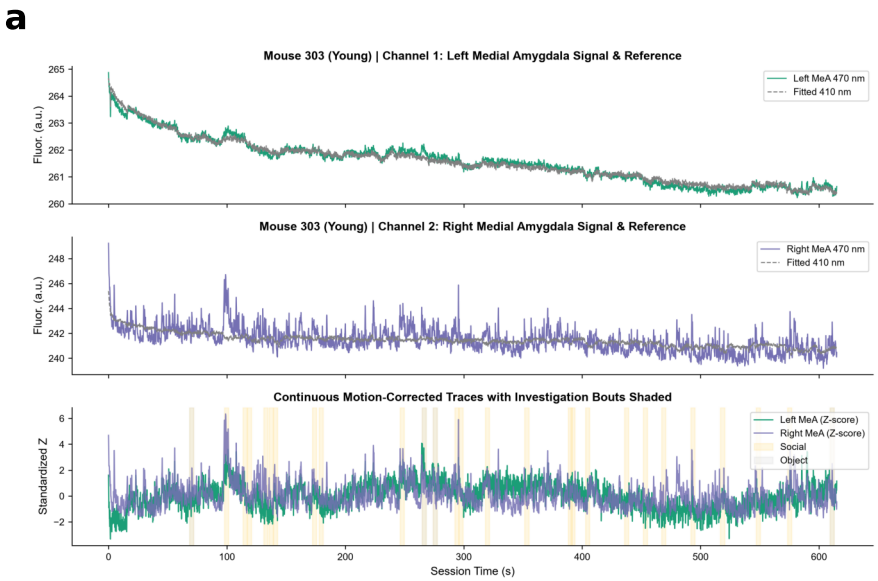
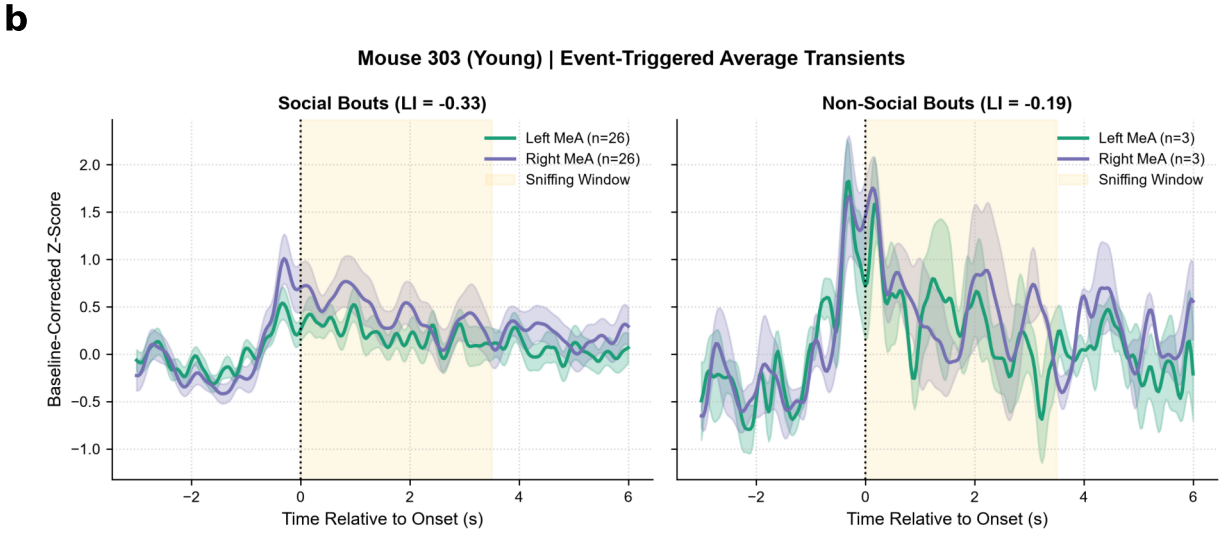
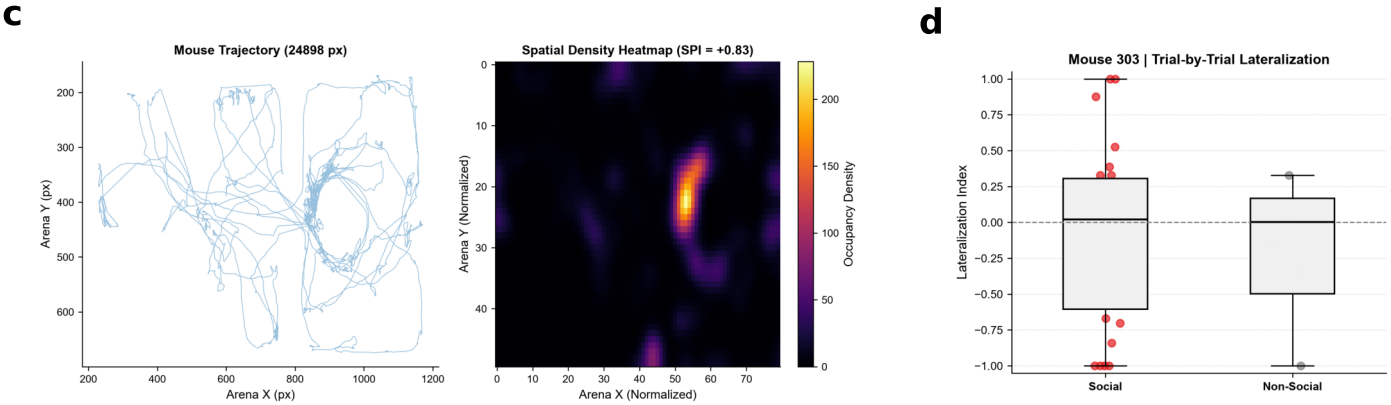
# Figure S1 | Bilateral Medial Amygdala Photometry & Spatial Behavior (Mouse 303)

Young C57BL/6 cohort; 3-chamber social preference test; 410 nm isosbestic motion correction




a, Raw fluorescence (470 nm, green/blue) and fitted 410 nm isosbestic reference (dashed grey) from left (CH1) and right (CH2) MeA, with motion-corrected Z-score trace (bottom). Shaded intervals denote social (yellow) and object (grey) investigation bouts. b, Event-aligned average traces (mean +/- SEM) around sniffing onset (t = 0 s, baseline: -3 to 0 s). c, Trajectory path and 2D spatial occupancy density map. d, Trial-by-trial lateralization index distribution. Summary metrics: 27 social bouts (mean duration: 3.6 s), Left MeA AUC = 0.81, Right MeA AUC = 1.60, Social LI = -0.33, Behavioral SPI = +0.83.